Design Calculation Sheet — T8-40g

VBF-T8R1-CALC-01 | Case t8_r1 — Long-endurance drone battery | 2026-08-25

Prepared: ________ Reviewed: ________ Approved: ________

Input parameters

Parameter	Value	Unit	Source
Positive current collector thickness	8	µm	r5_final_params.json (override; base 16)
Negative current collector thickness	6	µm	r5_final_params.json (override; base 12)
Positive particle radius	1	µm	r5_final_params.json (base 5.22)
Negative particle radius	1.5	µm	r5_final_params.json (base 5.86)
Electrolyte conductivity	1.3	S/m	r5_final_params.json (const override; lit. estimate)
Cation transference number	0.5	—	r5_final_params.json (base 0.2594)
Electrolyte diffusivity	3.5e-10	m²/s	r5_final_params.json (const override; lit. estimate)
Positive electrode porosity	0.45	—	r5_final_params.json
Negative electrode porosity	0.35	—	r5_final_params.json
Separator porosity	0.55	—	r5_final_params.json
Separator thickness	10	µm	r5_final_params.json (base 12)
Negative electrode thickness	95	µm	r5_final_params.json (base 85.2)
Total heat transfer coefficient	40	W/m²/K	r5_final_params.json (base 10)
Electrode height	0.072085	m	r5_final_params.json (scaled ×1.2299)
Electrode width	1.752222	m	r5_final_params.json (scaled ×1.2299)
Nominal cell capacity	6.149417	Ah	r5_final_params.json (5.0 × 1.2299)
Voltage window	2.5 – 4.2	V	chen2020_base_probe.json

Capacity and energy

Quantity	Value	Formula	Source
1C discharge capacity	6.942684712657194	∫ I_1C dt until 2.5 V (DFN)	r5_final_1c_dfn.json
1C discharge T_max	299.760756700537	lumped thermal DFN	r5_final_1c_dfn.json
5C discharge capacity	6.857050326442467	∫ I_5C dt until 2.5 V (DFN)	r5_final_5c_dfn.json
5C discharge T_max	319.9935140958404	lumped thermal DFN	r5_final_5c_dfn.json
5C capacity retention	0.987666	5C cap ÷ same-params 1C cap	r5_final_retention.json
Discharge energy (1C)	25.24595568575075	∫ V·I_1C dt / 3600	r5_final_energy.json
4C-45°C charge T_max	330.4128912521647	lumped thermal DFN, 4C CC to 4.2 V	r5_final_4c45_dfn.json
4C-45°C anode potential min	0.008174775130338882	min(anode_potential_v)	r5_final_4c45_dfn.json
4C charge acceptance	0.9395470466539817	∫ I_4C dt until 4.2 V cut-off	r5_final_4c45_dfn.json

Energy density

Quantity	Value	Unit	Formula	Source
Contract mass	0.03980000000000001	kg	Σ layer_th×(1−por)×density× area	r5_final_energy.json
Energy	25.24595568575075	Wh	∫ V·I_1C dt	r5_final_energy.json
Gravimetric ED	634.3204946168529	Wh/kg	energy ÷ mass	r5_final_energy.json
Stack volume	2.457973607824635e-05	m^3	Σ layer thickness × area	r5_final_energy.json
Volumetric ED	1027.104424774277	Wh/L	energy ÷ (volume×1000)	r5_final_energy.json
Midpoint voltage	3.875946280599504	V	V at time-midpoint of 1C discharge	r5_final_energy.json
DC internal resistance	0.0006840021584825283	Ω	(V■ − V_10%) ÷ I_1C	r5_final_energy.json
Power density	156142.2218592058	W/kg	V■^{2}/(4·R_DC)/mass	r5_final_energy.json

NP ratio and mass

Quantity	Value	Formula	Source
Positive capacity density (full)	70.32320803200001	Ah/m^2	(1−ε)·c_max·F/3600·th (window 0→1)
Negative capacity density (full)	54.83464331493056	Ah/m^2	(1−ε)·c_max·F/3600·th (window 0→1)
N/P (full-theoretical)	0.7797517327420344	—	neg density×th ÷ pos density×th
Positive capacity density (assembled)	51.33603101550001	Ah/m^2	full × (1−x■), x■=0.270 (probe)
Negative capacity density (assembled)	49.42780482430556	Ah/m^2	full × y■, y■=0.9014 (probe)
N/P (assembled-lithium basis)	0.9628287159438117	—	neg assembled ÷ pos assembled
Positive electrode mass	17.13179311432386	g	layer_kg_m2 × area × 1000
Negative electrode mass	12.92390776254958	g	layer_kg_m2 × area × 1000
Separator mass	0.2256510714480324	g	layer_kg_m2 × area × 1000
Al CC mass	2.728274919270921	g	layer_kg_m2 × area × 1000
Cu CC mass	6.790373132407625	g	layer_kg_m2 × area × 1000
Total mass (contract)	39.80000000000001	g	Σ layer masses (electrolyte excluded)
Electrolyte mass (lit. density)	11.02980921529694	g	pore volume × 1.2 g/cm^3 (annotated)

Process parameters

Parameter	Value	Formula	Source	
Positive areal density	135.63396	g/m^2	layer_kg_m2 × 1000	r5_final_energy.json
Negative areal density	102.31975	g/m^2	layer_kg_m2 × 1000	r5_final_energy.json
Positive compaction density	1.7941	g/cm^3	density×(1−ε) ÷ 1000	probe (density × 0.55)
Negative compaction density	1.07705	g/cm^3	density×(1−ε) ÷ 1000	probe (density × 0.65)
Electrolyte fill volume	9.191507679414116	cm^3	Σ(th×ε)×area×1e6	params + area
Electrolyte fill amount	11.02980921529694	g	pore vol × 1.2 g/cm^3 (lit.)	literature-annotated
Formation	0.1C CC to 4.2 V, 25 °C, 2 cycles	—	design recommended; production tuning required	